MEMORANDUM

TO: Dr. Tina Han

FROM: Hengeltraub, Isabella; Wu, Xintong; Jhanwar, Yuvraj; Hassan, Khaleel

CC: ME32301 Lab Section 08- Harbin, Zachary; Alshaikh, Leen

DATE: April-19-2025

RE: FEA and Design of 3D-Printed Parts (Design Project)

EXECUTIVE SUMMARY

The primary goal of this lab is to determine how infill density and filament color affect material properties such as yield strength σ_Y, ultimate tensile strength σ_{UTS}, Young's modulus E, and total strain at failure ϵ_F. Yield strength σ_Y of a material refers to the point at which elastic deformation of a material transitions to plastic deformation [1]. Ultimate tensile strength σ_{UTS} on the other hand is the maximum amount of stress a material can withstand before it breaks [2]. Young's modulus E refers to the ratio between stress and strain of a material. It is a measure of the ability of a material to withstand compression or elongation with respect to its length [3]. In addition, total failure at strain ϵ_F is the maximum strain the material experiences before it breaks [4]. The secondary objective of this lab is to attempt to model the Unfinished Purdue Block P to obtain the best strength-to-weight ratio possible. This was a virtual lab. The FEA results were provided along with the material properties database results containing the stress and strain values at different infill rates and filament colors. In the database provided, there were six different colors and six different infill densities. Prior to the start of data reduction and results analysis, the datasheet containing material properties and infill rates was sorted. Once the data had been properly organized, a graph between between infill density and material properties, namely σ_Y, σ_{UTS}, E, and ϵ_F, was plotted for each filament color. Additionally, the standard deviation for each data point was indicated in the graph to represent the inherent variability in material properties. From the plots, reasonable conclusions about the relationship between the parameters were drawn. The results showed that ϵ_F was highest

at 100% infill density, as more material can withstand higher strain [1]. σ_{UTS} shows a general trend such that it increases as infill density increases, which is consistent because greater material ensures specimen can bear more stress [2]. Lastly, σ_Y also increases as infill rate increases, because a higher infill density corresponds to a longer elastic deformation [3]. Since more material means greater stress and strain, E also increases as infill rate increases, as shown in [4]. Based on Lab 3-4, it was evident that the mode of failure of the 3D printed PLA material was in a ductile manner. As a result of this, the Maximum distortion-energy theorem was applied and the von Mises stresses were used to analyze the specimen. After analyzing the stress distribution on the Block, two sections that experienced the highest stress [5], were noted and their infill rates were increased to allow it to be able to withstand the stress better. More material was added to the weaker parts to reinforce the strength of the block. Furthermore, upon careful examination of the Unfinished Block P, it became obvious that there was a higher stress concentration factor around sharp edges of the specimen and the suggestion to fix this was to apply fillets around those edges (circled) [6] without compromising the shape or structural integrity of the specimen. Due to the inherent variability in the material properties, a safety factor of 1.2 was applied instead of 1. A factor of 1 may be too optimistic and fail to account for the potential inconsistencies introduced during the 3D printing process. However, since the component in question is subject to static loading, an excessively high safety factor was not necessary. Additionally, in regions where there was little stress experienced, material usage was decreased. From calculations performed, it was determined that the filament color that would best optimize the specific strength was blue. Utilizing infill densities of 25% and 40%, the total material usage came out to 9802.96 [7]. It is evident that the 3D printed material exhibited inherent variability, likely influenced by several factors associated with the printing process. Variables such as filament color, print time, filament type, nozzle and bed temperatures, ambient conditions, infill density and pattern, and layer thickness can all contribute to inconsistencies in the printed specimens. These variations directly impact key mechanical properties such as ultimate tensile stress, ultimate strength, Young's modulus, and total strain at failure, which must be carefully considered and accounted for during the design process.

RESULTS

Using the sample data, total strain at failure is plotted against the infill rate in Figure 1. The total strain at failure is generally highest at 100%, which makes sense because having more material allows the sample to stretch more. There is quite a bit of variability, which could mean that the 3D-printed part is less ductile than expected and breaks when there is less strain than anticipated.

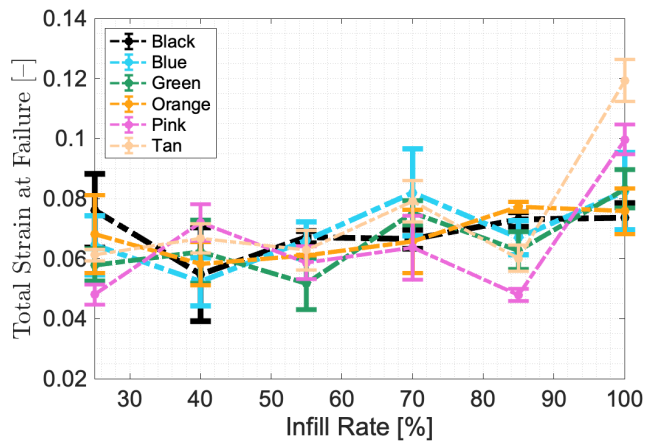

Figure 1: Total strain at failure of different infill rates

The ultimate tensile strength is plotted against the infill rate in Figure 2. As seen below, ultimate tensile strength generally increases as infill rate increases, which is expected because having more material allows the sample to withstand more stress. There is not a lot of variability that would cause the 3D-printed part to withstand less stress than anticipated.

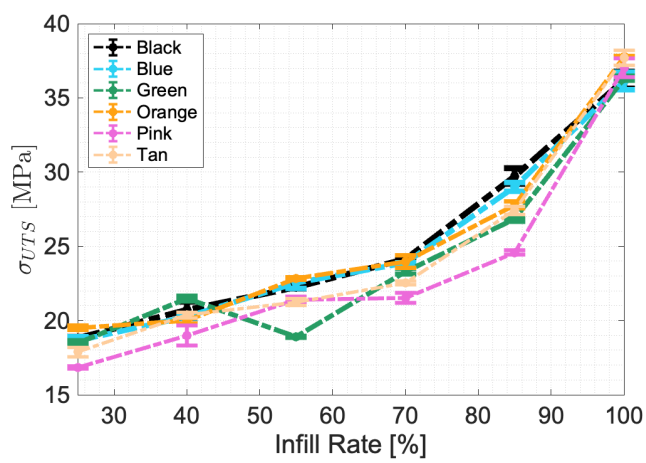

Figure 2: Ultimate tensile strength of different infill rates

Yield strength is plotted against the infill rate in Figure 3. Again, yield strength generally increases as infill rate increases, which makes sense because having more material allows the sample to have a larger elastic region. Again, there is quite a bit of variability, meaning the 3D-printed part could experience permanent deformation while under less stress than expected.

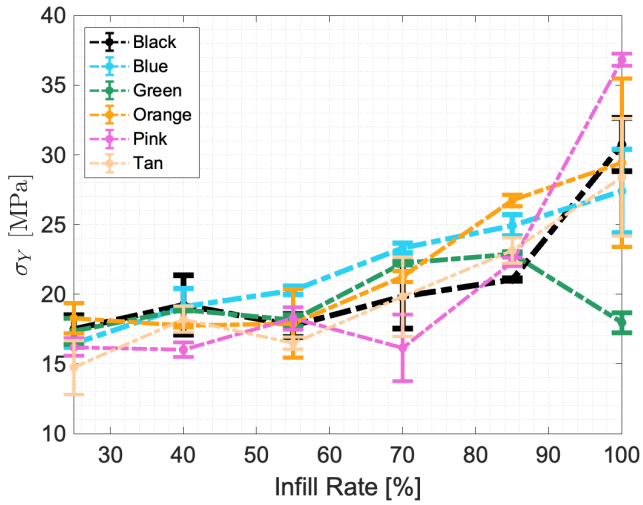

Figure 3: Yield strength of different infill rates

Young's modulus is plotted against the infill rate in Figure 4. Young's modulus also generally increases as infill rate increases, which is expected because having more material would make a sample stiffer. There is some variability, meaning the 3D-printed part may experience more or less strain than anticipated.

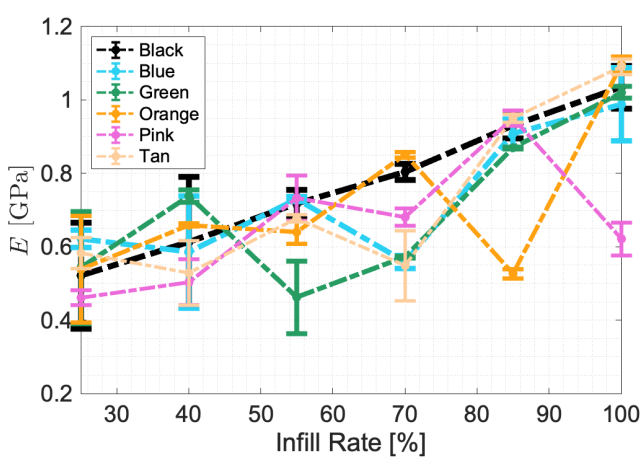

Figure 4: Young's modulus of different infill rates

A "for" loop was used to iterate through all the color and the subsequent infill rates for all the colors for the four figures above. It is important to note that the exhibited inherent variability is likely caused by several factors such as print time, ambient conditions, infill density and pattern, and layer thickness, among others, which are associated with the printing process. In turn, these variations affect mechanical properties of the material.

Since 3D-printed PLA behaves in a ductile manner under tension, the failure criteria is identified using the maximum-distortion-energy theory. Using the code file "BlockP.m", the Von Mises stress distribution was analyzed, as shown in Figure 5.

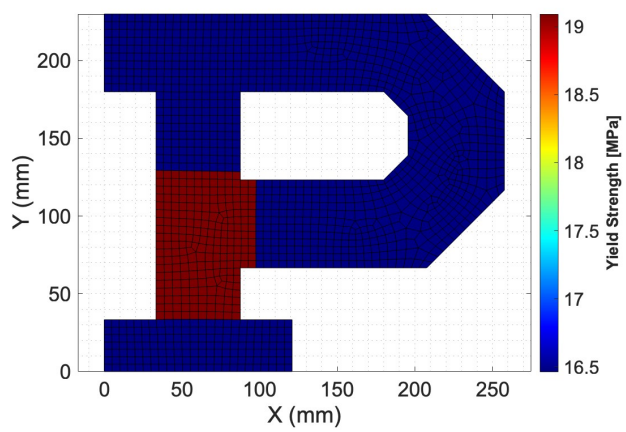

Figure 5: Von Mises stress distribution

The contour plot of proposed infill percentage is shown in Figure 6. As seen in the figure above, the stress distribution is most red in section 2—more precisely, 65mm-70mm. Because of this, section 2 was given a higher infill rate of 40%. Section 3 was given a 40% infill rate too, as the stress concentration in section 3 was considerably higher than the other sections. The geometric considerations advised are two corners in section 2, circled in red in the figure below. Having a fillet being introduced in those places could inherently reduce the sharp stress by spreading the stress over a larger area, making the design safer. This would also not involve greatly modifying the existing design, to maintain the honor of it. In addition, in regions where there was little stress experienced, material usage was decreased.

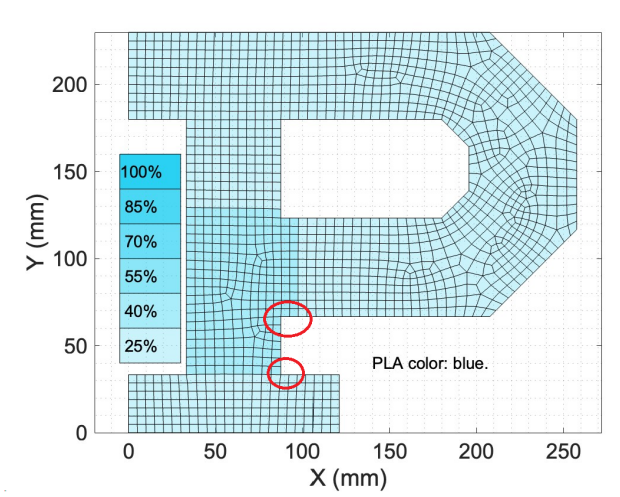

Figure 6: Contour plot of proposed infill percentage

Figure 7 shows the proposed color and infill rates, as well as the overall material usage for this design.

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Based on the PLA material properties gathered, we decide that:
The best PLA color for this design is blue;
Section 1 has the infill rate of  25%;
Section 2 has the infill rate of  40%;
Section 3 has the infill rate of  40%;
Section 4 has the infill rate of  25%;
Section 5 has the infill rate of  25%;
Section 6 has the infill rate of  25%;
Section 7 has the infill rate of  25%;
Overall Material usage is  9802.96. (The lower the better)
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Figure 7: Proposed design results

"CalculateMtrlUsage.m" was used to check which color and what order of infill rates would yield best results. The aim was to use the least amount of material while still ensuring that the 3D-printed part would not fail, so a factor of safety of a minimum of 1.2 was decided upon. This factor of safety was based on the general industry trend wherein an FS of 1.2 is an optimal value for a static lab-controlled design. Increasing strength in the design would require more material and therefore increase the weight, but increasing the infill rate generally causes the overall specific strength to decrease, meaning that the increase in strength is lower for the same amount of material at higher infill rates.

CONCLUSIONS

The main objective of this lab is to determine how 3D-printed filament density and color affect material properties such as yield strength, ultimate tensile strength, Young's Modulus, and total strain at failure. Then, that knowledge was applied using Finite Element Analysis, looking at stress distribution and points particularly prone to failure in the Purdue "Unfinished Block P," and manipulating the filament in this object to obtain the highest strength-to-weight ratio. This lab assumed that the 3D-printed materials were ductile, and thus used the Maximum- distortion-energy theory to analyze the Block P stress state.

The general trend was that material properties (yield strength, ultimate tensile strength, Young's Modulus, and total strain at failure) each increased as the infill rate increased. This trend was the clearest in Figures 2 and 3, for Ultimate Tensile Strength and Yield Strength respectively. Both of these properties increased for all colors as infill rate increased, with a steep slope from 90 to 100% infill. This makes sense, as having a higher infill rate (and thus more material) allows the sample to handle higher loads.

Figures 1 and 4, for Total Strain at Failure and Young's Modulus respectively, also observed this trend, although there was a lot more oscillation within each color filament, going up and down slightly between infill rates, with an overall increasing trend. Figures 1 and 4 observed much higher standard deviations than Figures 2 and 3, which may be related to this higher oscillation and lower linear trend.

Inherently, there is variability in 3D-printed materials, as can be seen in the standard deviation values of Figures 1-4, which comes from the layer-by-layer printing that happens in 3D printing, as there can be differences based on the printing parameters and setup. There can even be differences from machine to machine, due to parameters such as filament colors, print time, filament type, nozzle temperature, bed temperature, surrounding temperature, and infill densities, pattern and layer thickness. Due to this variability, it was important to make the entire "Unfinished Block P" one color and keep sections with the most similar infill percentages as possible, in order to avoid

further differences in section materials during printing.

A MATLAB code was used to decide on specific infill rates for the different sections of the design. This code was based on a factor of safety of 1.2, as that is the general industry trend for static lab-controlled design. The code used higher infill densities (more material) for regions of higher stress so that they would be able to withstand such stress, and lower infill densities (less material) for areas of lower stress; all of which would use the same color filament to ensure consistent infill properties (in this case, blue). As can be seen in Figure 5, Sections 2 and 3 have a higher stress distribution than the other sections, and as such, they were assigned infill rates of 40%, as opposed to 25% infill for the rest of the sections, which demonstrated a lower stress distribution. As such, the team was able to optimize material to be as low as 9802.96, a can be seen in Figure 7.

This was done to ensure that Block P would be able to withstand Von Mises stress, which is the stress considered in failure theories for ductile materials. This was done while using the least amount of materials in order to achieve an optimal strength-to-weight ratio. Higher infill percentages allow for overall higher specific strength of the block but also increase weight, so there needed to be a trade-off where more material was used only for the sections that needed it.

On top of changing infill distribution based on the FEA results seen in Figure 5, geometric considerations were made for ”Unfinished Block P.” As can be seen in Figure 6, the team advises that fillets be introduced in the two circled sections, as having those sharp corners reduced would allow for the stress in those areas to be spread over a larger area, making the design safer and less likely to break.

If the design were to be 3D printed and tested in axial tension, as it would be subjected to during lifting and placement, with the discussed calculated section infill rates and geometric adjustments, the block would not be expected to fail. However, if it were to be loaded at a higher axial stress, or a compressive stress, it would most likely crack along the circled areas in Figure 6, as well as on the bottom left corner of Section 2, as these sharper corners would make for higher stress concentrations and higher chances of ultimate failure and fracture.

REFERENCE LIST

[1] X. RSS. "Learn all about yield strength." (2024), [Online]. Available: https://www.xometry.com/resources/3d-printing/yield-strength/#:~:text=Yield%20strength%20tells%20engineers%20when,its%20original%20shape%20or%20form. (visited on 04/18/2025).

[2] M. Supermarkets. "What is ultimate tensile strength?" (2023), [Online]. Available: https://www.metalsupermarkets.com/what-is-ultimate-tensile-strength/#:~:text=Ultimate%20tensile%20strength%20(UTS)%2C,component%20will%20behave%20under%20load. (visited on 04/18/2025).

[3] E. Britannica. "Young's modulus." (2025), [Online]. Available: https://www.britannica.com/science/Youngs-modulus (visited on 04/18/2025).

[4] MechaniCalc. "Mechanical properties of materials." (2025), [Online]. Available: https://mechanicalc.com/reference/mechanical-properties-of-materials (visited on 04/18/2025).

[5] OpenAI. "ChatGPT (April 18th, 2025 Version)," [Online]. Available: https://openai.com/chatgpt(visitedon18/04/2025).

DATA SHEETS

The following four images are screenshots of the updated dataset containing various material properties such as Young's Modulus [E], Yield Strength, Ultimate Tensile Strength, and Total Strain at Failure for 6 different colours and 6 different infill rates. The actual dataset contained the same information but in random order of colors and infill rates. The following images contain tables produced after running code to sort the datasets such that each color and its infill rates are displayed in ascending order. This was a key step in ensuring that the plots [] could be generated more efficiently.

1	Color	Infill [%]	Youngs Modulus Mean [GPa]	Youngs Modulus Sdv [GPa]
2	black	25	0.5214753566	0.1444926494
3	black	40	0.6135857062	0.1760866942
4	black	55	0.7170359426	0.03820901464
5	black	70	0.803214428	0.02193714712
6	black	85	0.9301623943	0.03564705255
7	black	100	1.034558119	0.05824975862
8	blue	25	0.6214283606	0.02331316505
9	blue	40	0.5853750971	0.1534610523
10	blue	55	0.7293105952	0.006611674109
11	blue	70	0.5587079704	0.01752748739
12	blue	85	0.9086555744	0.03933793228
13	blue	100	0.9880813485	0.09900596691
14	green	25	0.5430674901	0.1522108274
15	green	40	0.7379821065	0.01611553201
16	green	55	0.4619341012	0.0987246598
17	green	70	0.5719443301	0.002689773491
18	green	85	0.8695734108	0.002551733275
19	green	100	1.020918371	0.01550451098
20	orange	25	0.5384962294	0.1448922573
21	orange	40	0.6592050876	0.003876339755
22	orange	55	0.6398931905	0.0313234411
23	orange	70	0.8509527631	0.007817516183
24	orange	85	0.5262953922	0.01311468081
25	orange	100	1.096041316	0.02204280846
26	pink	25	0.4613234762	0.02046903986
27	pink	40	0.5033917785	0.06283301095
28	pink	55	0.7320999481	0.06242796745
29	pink	70	0.6807436727	0.023290238
30	pink	85	0.9501654767	0.02143883915
31	pink	100	0.6207985102	0.04449838908
32	tan	25	0.5825717957	0.04171594369
33	tan	40	0.5288781997	0.08825022781
34	tan	55	0.6768251028	0.01012731
35	tan	70	0.5486865974	0.09531045239

Figure 8: Dataset for Youngs Modulus (GPa) of Different Colors at Varying Infill Rates

1	Color	Infill [%]	Yield Strength Mean [MPa]	Yield Strength Sdv [MPa]
2	black	25	17.49795159	1.000705781
3	black	40	19.23658124	2.126363779
4	black	55	17.75064475	0.8120786293
5	black	70	19.83851144	2.311574695
6	black	85	21.04740893	0.07301584167
7	black	100	30.72236662	1.909097128
8	blue	25	16.46619117	0.28553658
9	blue	40	19.09176539	1.291133786
10	blue	55	20.28744056	0.3107625456
11	blue	70	23.3266844	0.352613089
12	blue	85	24.93225137	0.7568715295
13	blue	100	27.39695408	3.009168603
14	green	25	17.35743825	0.9329902108
15	green	40	18.88461577	0.1095237625
16	green	55	18.12653631	0.4140532485
17	green	70	22.23741739	0.1389081866
18	green	85	22.85886626	0.1736352332
19	green	100	17.94960912	0.7355010392
20	orange	25	18.28237075	1.087876252
21	orange	40	17.75001511	0.1344560011
22	orange	55	17.89294205	2.450482881
23	orange	70	21.28037355	0.4006966921
24	orange	85	26.72544931	0.3989158179
25	orange	100	29.44232048	6.022026063
26	pink	25	16.20552265	0.6215250913
27	pink	40	16.01789168	0.5200152626
28	pink	55	18.26001874	0.8147499405
29	pink	70	16.13122582	2.393494907
30	pink	85	22.30478824	0.3161051682
31	pink	100	36.82478038	0.4185054339
32	tan	25	14.74193816	1.920488584
33	tan	40	18.20901838	0.95276769
34	tan	55	16.49578397	0.4532324806
35	tan	70	19.82528913	2.834261269

Figure 9: Dataset for Yield Strength (MPa) of Different Colors at Varying Infill Rates

1	Color	Infill [%]	Ultimate Tensile Strength Mean [MPa]	Ultimate Tensile Strength Sdv [MPa]
2	black	25	18.80381273	0.1152373766
3	black	40	20.723777	0.5307005077
4	black	55	22.26260276	0.09972895447
5	black	70	24.12317154	0.08102977551
6	black	85	29.73572997	0.5440570641
7	black	100	36.21088713	0.5654275544
8	blue	25	18.64661408	0.2690800124
9	blue	40	20.24966252	0.0685636562
10	blue	55	22.51949347	0.3668600825
11	blue	70	23.88202168	0.02137049024
12	blue	85	29.00472477	0.3080912343
13	blue	100	36.14414591	0.5800881595
14	green	25	18.49203891	0.147903272
15	green	40	21.49256024	0.1665117365
16	green	55	18.90476406	0.04897404014
17	green	70	23.31409172	0.1852109154
18	green	85	26.83500564	0.1406890608
19	green	100	36.32359163	0.1371273124
20	orange	25	19.50512018	0.1490318794
21	orange	40	19.98773473	0.2021292202
22	orange	55	22.83808833	0.08904370935
23	orange	70	23.9840224	0.4202863081
24	orange	85	27.78197534	0.2279518959
25	orange	100	37.76734264	0.06678278201
26	pink	25	16.84460127	0.07746802713
27	pink	40	18.98220906	0.6802939394
28	pink	55	21.41322635	0.1985674718
29	pink	70	21.51585671	0.3258999762
30	pink	85	24.58469334	0.1282229415
31	pink	100	37.03318927	0.6259772767
32	tan	25	17.87247895	0.3280216445
33	tan	40	20.36551519	0.1878822267
34	tan	55	21.24574367	0.2306232072
35	tan	70	22.53460469	0.1406890608

Figure 10: Dataset for Ultimate Tensile Strength (MPa) of Different Colors at Varying Infill Rates

1	Color	Infill [%]	Total Strain at Failure Mean [-]	Total Strain at Failure Sdv [-]
2	black	25	0.07600310559	0.01225281475
3	black	40	0.05477484472	0.01569689215
4	black	55	0.06715295031	0.001988463324
5	black	70	0.06642546584	0.003188568467
6	black	85	0.07301708075	0.002488049637
7	black	100	0.07361723602	0.004876182011
8	blue	25	0.06446467391	0.009846506896
9	blue	40	0.05227406832	0.007992283013
10	blue	55	0.06610636646	0.006156440562
11	blue	70	0.08198059006	0.01444188741
12	blue	85	0.06699145963	0.005690891998
13	blue	100	0.08241964286	0.01289350737
14	green	25	0.05789013975	0.0053697395
15	green	40	0.06232375776	0.01056158405
16	green	55	0.05164984472	0.008604962491
17	green	70	0.07517468944	0.004035120219
18	green	85	0.06267779503	0.006261847784
19	green	100	0.08321273292	0.006414468658
20	orange	25	0.0680951087	0.01296611531
21	orange	40	0.05828881988	0.007169887083
22	orange	55	0.06114596273	0.0001515228817
23	orange	70	0.06567080745	0.01057366196
24	orange	85	0.07722981366	0.001583304314
25	orange	100	0.07576630435	0.007662885444
26	pink	25	0.04797903727	0.003418048773
27	pink	40	0.07195807453	0.006133382732
28	pink	55	0.05863897516	0.005486665506
29	pink	70	0.06359704969	0.01070212701
30	pink	85	0.04792546584	0.002040068943
31	pink	100	0.09960326087	0.004959629395
32	tan	25	0.06116925466	0.00185952289
33	tan	40	0.06666614907	0.004870692052
34	tan	55	0.06279968944	0.00653744375
35	tan	70	0.07912034161	0.006929426857

Figure 11: Dataset for Total Strain at Failure (-) of Different Colors at Varying Infill Rates

APPENDIX

The following code was used to put the dataset given to us, in ascending order of infll rates (25, 40, 55, 70, 85, 100) and in the order of black, blue, green, orange, pink, tan.

```
clc
clear all
close all

%% Read Table of all sample data
% Do not change.
data=readtable("2025_SP_SampleData.xlsx",'VariableNamingRule','preserve');

%% Gather a list of color and infill rate
% Do not change.
color_vec=unique(data.Color);
infill_vec=unique(data.("Infill[%]"));

%% Initialize the table for the summary
% Do not change
% Calculate summary table size
sz = [length(color_vec)*length(infill_vec) 10];
% Set up summary table heading
varTypes = ["string","int16", "double", "double","double","double",...
            "double", "double","double","double"];
varNames = ["Color", "Infill[%]",...
            "Youngs Modulus Mean[GPa]","Youngs Modulus Sdv[GPa]",...
            "Yield Strength Mean[MPa]","Yield Strength Sdv[MPa]"...
            "Ultimate Tensile Strength Mean[MPa]","Ultimate Tensile Strength Sdv[MPa]"...
            "Total Strain at Failure Mean[-]", "Total Strain at Failure Sdv[-]"];

```

```
SampleDataSummary=table('Size',sz,'VariableNames',varNames,'VariableTypes',
    varTypes);

%% Gather data by color and infill rate for calculation
% Do not change
counter=1;

for ii=1:length(color_vec) % Loop through all possible color
    for jj=1:length(infill_vec) % Loop through all possible infill rate

        temp_color=color_vec(ii);
        temp_infill=infill_vec(jj);

        temp_data_infill=data(data.("Infill[%]")==temp_infill,:);

        temp_color_check=zeros(height(temp_data_infill),1);
        for kk=1:height(temp_data_infill)
            if isequal(string(temp_data_infill.Color{kk}), temp_color)
                temp_color_check(kk)=1;
            end
        end

        if sum(temp_color_check)==0 % output warning message if a combination
    of infill rate and color doesnot exist
                fprintf("Error: No Sample is %s with the infill rate of %3.0f
    .\n", string(temp_color), temp_infill);
            end

% This is all the specimens that has the current infillrate and color
        temp_data=temp_data_infill(logical(temp_color_check),:);
% If you have issue understading what is temp_data, uncomment the following
% line in Matlab Debuging mode.
        %disp(temp_data)a
```

```

%% Calculate mean and SDV of material properties.
% currenty all of these values are set to 1 so the code could run but they
% need to be edited to calculate mean and SDV of each material property.

        temp_E_mean= mean(temp_data.("Youngs Modulus[GPa]"));
        temp_E_sdv= std(temp_data.("Youngs Modulus[GPa]"));
        temp_sigY_mean= mean(temp_data.("Yield Strength[MPa]"));
        temp_sigY_sdv= std(temp_data.("Yield Strength[MPa]"));
        temp_sigUTS_mean= mean(temp_data.("Ultimate Tensile Strength[MPa]"));
        temp_sigUTS_sdv= std(temp_data.("Ultimate Tensile Strength[MPa]"));
        temp_epsF_mean= mean(temp_data.("Total Strain at Failure[-]"));
        temp_epsF_sdv= std(temp_data.("Total Strain at Failure[-]"));

%% Enter the Calculated property mean and sdv into the summary table
% Do not change
        SampleDataSummary(counter,:)={temp_color, temp_infill,...
                                      temp_E_mean, temp_E_sdv,...
                                      temp_sigY_mean, temp_sigY_sdv,...
                                      temp_sigUTS_mean, temp_sigUTS_sdv,...
                                      temp_epsF_mean,temp_epsF_sdv};
%% Move on to next line
% Do not change
        counter=counter+1;


    end
end

%% Ording the Summary table so they arein in the correct order and save as
    file.
% Do not change
SampleDataSummary=sortrows(SampleDataSummary,[1,2],{'ascend','ascend'});
```

```
writetable(SampleDataSummary,'SampleDataSummary.xlsx');
```

The following code was used to generate the 4 plots that show the trend for Youngs Modulus (E), Yield Strength (σ_Y), Ultimate Tensile Strength (σ_{UTS}), Total Strain at Failure (ϵ_F), with increasing infill rates.

```
clc
clear all
close all

%% Read Table of Data Sumamry
% Do not change.
data=readtable("SampleDataSummary.xlsx",'VariableNamingRule','preserve');
color_vec=unique(data.Color);
infill_vec = [25,40,55,70,85,100];

%% Preset of 6 colors that match with the filament color, use for plots.
% Do not change
% Black Blue Green Orange Pink Tan
colorlist=["#000000","#2bd1f3", "#269c63","#ffa111","#ec6ada", "#fecd9c"];


%% Generating figures for 4 material properties

% Currently only generating graph for sigma_Y. Need to edit so it can
% generate 4 figures at once.
for ii=1:length(color_vec)

    % index numbers of the rows with one color's 6 infill rate
    % Need edit, currently set to row 1 through 6.
    ind=[ii*6-5:ii*6];

    % Need edit to read in mean and SDV of the color's sigma_Y value,
```

```
    % currently set to a vector of [1 2 3 4 5 6];
    temp_sigY_mean=data.("Yield Strength Mean[MPa]")(ind);
    temp_sigY_sdv=data.("Yield Strength Sdv[MPa]")(ind);


    % Uncomment and the following lines after making sure the sigma_Y plot
    % is correct comparing to the one provided in the prelab lecture slides
    temp_E_mean=data.("Youngs Modulus Mean[GPa]")(ind);
    temp_E_sdv=data.("Youngs Modulus Sdv[GPa]")(ind);
    temp_sigUTS_mean=data.("Ultimate Tensile Strength Mean[MPa]")(ind);
    temp_sigUTS_sdv=data.("Ultimate Tensile Strength Sdv[MPa]")(ind);
    temp_epsF_mean=data.("Total Strain at Failure Mean[-]")(ind);
    temp_epsF_sdv=data.("Total Strain at Failure Sdv[-]")(ind);


    % Plot the graph of sigma_y with errorbar()
    % adjust as necessary
    figure(1)
    hold on
    p_sdv=errorbar(infill_vec,temp_sigY_mean,temp_sigY_sdv);
    p_sdv.LineStyle="-."; % dot dash style line fore visibility
    p_sdv.LineWidth=5-ii*0.4; % line thickness reduces with each iteration for
     visibility
    p_sdv.Color=colorlist(ii); % set each line color the same as filament
   color
    p_sdv.Marker='.';
    p_sdv.MarkerSize=25;
    p_sdv.CapSize=15; %set error bar size

    % Need to edit to show the other 3 material properties after checking
    % sigma_Y graph looks correct
    figure(2)
    hold on
```

```
    p_sdv=errorbar(infill_vec,temp_E_mean,temp_E_sdv);
    p_sdv.LineStyle="-."; % dot dash style line fore visibility
    p_sdv.LineWidth=5-ii*0.4; % line thickness reduces with each iteration for visibility
    p_sdv.Color=colorlist(ii); % set each line color the same as filament color
    p_sdv.Marker='.';
    p_sdv.MarkerSize=25;
    p_sdv.CapSize=15; %set error bar size


    figure(3)
    hold on
    p_sdv=errorbar(infill_vec,temp_sigUTS_mean,temp_sigUTS_sdv);
    p_sdv.LineStyle="-."; % dot dash style line fore visibility
    p_sdv.LineWidth=5-ii*0.4; % line thickness reduces with each iteration for visibility
    p_sdv.Color=colorlist(ii); % set each line color the same as filament color
    p_sdv.Marker='.';
    p_sdv.MarkerSize=25;
    p_sdv.CapSize=15; %set error bar size

    figure(4)
    hold on
    p_sdv=errorbar(infill_vec,temp_epsF_mean,temp_epsF_sdv);
    p_sdv.LineStyle="-."; % dot dash style line fore visibility
    p_sdv.LineWidth=5-ii*0.4; % line thickness reduces with each iteration for visibility
    p_sdv.Color=colorlist(ii); % set each line color the same as filament color
    p_sdv.Marker='.';
    p_sdv.MarkerSize=25;
```

```
    p_sdv.CapSize=15; %set error bar size

end

%% Adjust legend setting for figure 1
% Edit as necessary
figure(1)
box on
grid minor
legend('Black','Blue','Green','Orange','Pink','Tan', ...
       'Location','northwest','Color','w', ...
       'FontName','Arial','FontSize',14);
xlim([25 100]);
xlabel('Infill Rate [%]');
ylabel('$\sigma_Y$ [MPa]','interpreter','latex')
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;
set(figure(1), 'ToolBar', 'none');
set(gcf, 'Units', 'inches', 'Position', [1 1 7.5 5]);
exportgraphics(gcf,'25SPLab10YieldStrength.png');

%% Adjust legend and axis setting for other 3 figures
% Needs edit.
figure(2)
box on
grid minor
legend('Black','Blue','Green','Orange','Pink','Tan', ...
       'Location','northwest','Color','w', ...
       'FontName','Arial','FontSize',14);
xlim([25 100]);
xlabel('Infill Rate [%]');
ylabel('$E$ [GPa]','interpreter','latex')
```

```
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;
set(figure(2), 'ToolBar', 'none');
set(gcf, 'Units', 'inches', 'Position', [1 1 7.5 5]);
exportgraphics(gcf,'25SPLab10Youngs.png');

figure(3)
box on
grid minor
legend('Black','Blue','Green','Orange','Pink','Tan', ...
       'Location','northwest','Color','w', ...
       'FontName','Arial','FontSize',14);
xlim([25 100]);
xlabel('Infill Rate [%]');
ylabel('$\sigma_{UTS}$ [MPa]','interpreter','latex')
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;
set(figure(3), 'ToolBar', 'none');
set(gcf, 'Units', 'inches', 'Position', [1 1 7.5 5]);
exportgraphics(gcf,'25SPLab10UTS.png');


figure(4)
box on
grid minor
legend('Black','Blue','Green','Orange','Pink','Tan', ...
       'Location','northwest','Color','w', ...
       'FontName','Arial','FontSize',14);
xlim([25 100]);
xlabel('Infill Rate [%]');
ylabel('Total Strain at Failure [--]','interpreter','latex')
```

```
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;
set(figure(4), 'ToolBar', 'none');
set(gcf, 'Units', 'inches', 'Position', [1 1 7.5 5]);
exportgraphics(gcf,'25SPLab10TotStrainatFailure.png');
```

The following code was used to generate the command window output that shows the infill rates used per section on the Unfinished P block model along with the total material usage. Furthermore, this code also provided the stress contour plots.

```
clc
clear all
close all

%% Input area

% choose from one of the folloing six:
% Black Blue Green Orange Pink Tan
PLA_color='blue';

% Infill rate of each of each section:
% should be a 1x7 vector with each entry being one of the following
% 25 40 55 70 85 100
infill_rate_sec=[25 40 40 25 25 25 25];

%% DO NOT CHANGE THE REST OF THE CODE!!!

%%Black Blue Green Orange Pink Tan
colornames={'black' 'blue' 'green' 'orange' 'pink' 'tan'};
colorlist=["#000000","#2bd1f3", "#269c63","#ffa111","#ec6ada", "#fecd9c"];

%% Set PLA infill color
```

```
sec_facealpha=infill_rate_sec./100;

if strcmpi(PLA_color,colornames(1)); fill_color=colorlist(1);
else if strcmpi(PLA_color,colornames(2));fill_color=colorlist(2);
else if strcmpi(PLA_color,colornames(3));fill_color=colorlist(3);
else if strcmpi(PLA_color,colornames(4));fill_color=colorlist(4);
else if strcmpi(PLA_color,colornames(5));fill_color=colorlist(5);
else if strcmpi(PLA_color,colornames(6));fill_color=colorlist(6);
else fprintf('Error!\nInvalid choice of PLA color!\n'); return;
end; end; end; end; end; end


%% Check infill rate
valid_infill_rate=[25 40 55 70 85 100];
if any(not(ismember(infill_rate_sec,valid_infill_rate)))
    fprintf('Error!\nInvalid choice of infill rate!\n'); return;
end

%%
MatProp=readtable("SampleDataSummary.xlsx",'VariableNamingRule','preserve');
idx_1=string(MatProp.Color)==lower(PLA_color)...
      & MatProp.("Infill[%]")==infill_rate_sec(1);
sec_1_sigY=MatProp.("Yield Strength Mean[MPa]")(idx_1);
idx_2=string(MatProp.Color)==lower(PLA_color)...
      & MatProp.("Infill[%]")==infill_rate_sec(2);
sec_2_sigY=MatProp.("Yield Strength Mean[MPa]")(idx_2);
idx_3=string(MatProp.Color)==lower(PLA_color)...
      & MatProp.("Infill[%]")==infill_rate_sec(3);
sec_3_sigY=MatProp.("Yield Strength Mean[MPa]")(idx_3);
idx_4=string(MatProp.Color)==lower(PLA_color)...
      & MatProp.("Infill[%]")==infill_rate_sec(4);
sec_4_sigY=MatProp.("Yield Strength Mean[MPa]")(idx_4);
idx_5=string(MatProp.Color)==lower(PLA_color)...
```

```
      & MatProp.("Infill[%]")==infill_rate_sec(5);
sec_5_sigY=MatProp.("Yield Strength Mean[MPa]")(idx_5);
idx_6=string(MatProp.Color)==lower(PLA_color)...
      & MatProp.("Infill[%]")==infill_rate_sec(6);
sec_6_sigY=MatProp.("Yield Strength Mean[MPa]")(idx_6);
idx_7=string(MatProp.Color)==lower(PLA_color)...
      & MatProp.("Infill[%]")==infill_rate_sec(7);
sec_7_sigY=MatProp.("Yield Strength Mean[MPa]")(idx_7);

sec_sigY=[sec_1_sigY sec_2_sigY sec_3_sigY sec_4_sigY...
          sec_5_sigY sec_6_sigY sec_7_sigY];


%%z
Mesh = readtable('BlockPMesh.mphtxt', 'FileType', 'text');
% Reading in Nodes From Mesh
n_node= Mesh{13,1};
mesh_nodes = Mesh{16:15+n_node,1:2};

% Reading in Elements
n_elements = Mesh{13+7+n_node,1};
con_matrix_p1 = Mesh(13+9+n_node : 13+8+n_node+n_elements, 1:2);
con_matrix_p2 = str2double(table2array(Mesh(13+9+n_node : 13+8+n_node+
    n_elements, 3:4)));
con_matrix= table2array([con_matrix_p1, array2table(con_matrix_p2)]) + 1;
con_matrix(:, [1, 2]) = con_matrix(:, [2, 1]);

% Read Nodes Coordinates for Seciton 1-6
sec_1_nodes_xyz=readmatrix('sec_1_nodes_xyz.xyz','FileType','text');
sec_1_nodes_xy=sec_1_nodes_xyz(:,1:2);
sec_2_nodes_xyz=readmatrix('sec_2_nodes_xyz.xyz','FileType','text');
sec_2_nodes_xy=sec_2_nodes_xyz(:,1:2);
sec_3_nodes_xyz=readmatrix('sec_3_nodes_xyz.xyz','FileType','text');
```

```
sec_3_nodes_xy=sec_3_nodes_xyz(:,1:2);
sec_4_nodes_xyz=readmatrix('sec_4_nodes_xyz.xyz','FileType','text');
sec_4_nodes_xy=sec_4_nodes_xyz(:,1:2);
sec_5_nodes_xyz=readmatrix('sec_5_nodes_xyz.xyz','FileType','text');
sec_5_nodes_xy=sec_5_nodes_xyz(:,1:2);
sec_6_nodes_xyz=readmatrix('sec_6_nodes_xyz.xyz','FileType','text');
sec_6_nodes_xy=sec_6_nodes_xyz(:,1:2);


sec_nodes_dis=zeros(height(mesh_nodes),6);
sec_nodes_check=zeros(height(mesh_nodes),6);


tol=1;
for ii=1:height(mesh_nodes)
    temp_node=mesh_nodes(ii,:);

    sec_1_dist_sq=min(sum((temp_node-sec_1_nodes_xy).^2,2));
    if sec_1_dist_sq<=tol; sec_nodes_check(ii,1)=1; end

    sec_2_dist_sq=min(sum((temp_node-sec_2_nodes_xy).^2,2));
    if sec_2_dist_sq<=tol; sec_nodes_check(ii,2)=1; end

    sec_3_dist_sq=min(sum((temp_node-sec_3_nodes_xy).^2,2));
    if sec_3_dist_sq<=tol; sec_nodes_check(ii,3)=1; end

    sec_4_dist_sq=min(sum((temp_node-sec_4_nodes_xy).^2,2));
    if sec_4_dist_sq<=tol; sec_nodes_check(ii,4)=1; end

    sec_5_dist_sq=min(sum((temp_node-sec_5_nodes_xy).^2,2));
    if sec_5_dist_sq<=tol; sec_nodes_check(ii,5)=1; end

    sec_6_dist_sq=min(sum((temp_node-sec_6_nodes_xy).^2,2));
```

```
    if sec_6_dist_sq<=tol; sec_nodes_check(ii,6)=1; end

end


%%
sec_1_nodes_ind=find(sec_nodes_check(:,1)>0);
sec_1_is_valid = all(ismember(con_matrix, sec_1_nodes_ind), 2);
sec_1_elements = con_matrix(sec_1_is_valid, :);

sec_2_nodes_ind=find(sec_nodes_check(:,2)>0);
sec_2_is_valid = all(ismember(con_matrix, sec_2_nodes_ind), 2);
sec_2_elements = con_matrix(sec_2_is_valid, :);

sec_3_nodes_ind=find(sec_nodes_check(:,3)>0);
sec_3_is_valid = all(ismember(con_matrix, sec_3_nodes_ind), 2);
sec_3_elements = con_matrix(sec_3_is_valid, :);

sec_4_nodes_ind=find(sec_nodes_check(:,4)>0);
sec_4_is_valid = all(ismember(con_matrix, sec_4_nodes_ind), 2);
sec_4_elements = con_matrix(sec_4_is_valid, :);

sec_5_nodes_ind=find(sec_nodes_check(:,5)>0);
sec_5_is_valid = all(ismember(con_matrix, sec_5_nodes_ind), 2);
sec_5_elements = con_matrix(sec_5_is_valid, :);

sec_6_nodes_ind=find(sec_nodes_check(:,6)>0);
sec_6_is_valid = all(ismember(con_matrix, sec_6_nodes_ind), 2);
sec_6_elements = con_matrix(sec_6_is_valid, :);

sec_7_not_valid=sec_1_is_valid+sec_2_is_valid+sec_3_is_valid...
               +sec_4_is_valid+sec_5_is_valid+sec_6_is_valid;
sec_7_is_valid=not(sec_7_not_valid);
```

```
sec_7_elements = con_matrix(sec_7_is_valid,:);


%% Generate Grid
legend_nodes=[-5 40; 30 40;-5 60; 30 60;-5 80; 30 80;-5 100; 30 100;...
              -5 120; 30 120;-5 140; 30 140;-5 160; 30 160;];
legend_ele=[1 2 4 3; 3 4 6 5; 5 6 8 7; 7 8 10 9; 9 10 12 11;...
            11 12 14 13];

figure (2)
hold on;
box on
grid minor

% Plotting the quad mesh using patch
hPatch_sec_1= patch('Faces', sec_1_elements, 'Vertices', mesh_nodes,  ...
      'FaceColor', fill_color, 'EdgeColor', 'black', ...
      'FaceAlpha',sec_facealpha(1), 'FaceLighting', 'none');

hPatch_sec_2= patch('Faces', sec_2_elements, 'Vertices', mesh_nodes,  ...
      'FaceColor', fill_color, 'EdgeColor', 'black', ...
      'FaceAlpha',sec_facealpha(2), 'FaceLighting', 'none');

hPatch_sec_3= patch('Faces', sec_3_elements, 'Vertices', mesh_nodes,  ...
      'FaceColor', fill_color, 'EdgeColor', 'black', ...
      'FaceAlpha',sec_facealpha(3), 'FaceLighting', 'none');

hPatch_sec_4= patch('Faces', sec_4_elements, 'Vertices', mesh_nodes,  ...
      'FaceColor', fill_color, 'EdgeColor', 'black', ...
      'FaceAlpha',sec_facealpha(4), 'FaceLighting', 'none');

hPatch_sec_5= patch('Faces', sec_5_elements, 'Vertices', mesh_nodes,  ...
      'FaceColor', fill_color, 'EdgeColor', 'black', ...
```

```
      'FaceAlpha',sec_facealpha(5), 'FaceLighting', 'none');

hPatch_sec_6= patch('Faces', sec_6_elements, 'Vertices', mesh_nodes,  ...
      'FaceColor', fill_color, 'EdgeColor', 'black', ...
      'FaceAlpha',sec_facealpha(6), 'FaceLighting', 'none');

hPatch_sec_7= patch('Faces', sec_7_elements, 'Vertices', mesh_nodes,  ...
      'FaceColor', fill_color, 'EdgeColor', 'black', ...
      'FaceAlpha',sec_facealpha(7), 'FaceLighting', 'none');

for iii=1:6
    hPatch_legend=patch('Faces', legend_ele(iii,:), 'Vertices', legend_nodes,
    ...
        'FaceColor', fill_color, 'EdgeColor', 'black', ...
        'FaceAlpha',0.1+iii*0.15, 'FaceLighting', 'none');
    text_box=sprintf('%3.0f%%',iii*15+10);
    Annodataion=text(-5,30+iii*20,text_box,'FontName','Arial','FontSize',14);
end
text_box=sprintf('PLA color: %s.',PLA_color);
Annodataion=text(140,40,text_box,'FontName','Arial','FontSize',14);

sec_1_area=section_area_quad_mesh(hPatch_sec_1);

% Set axis properties
axis equal;
xlabel('X (mm)');
ylabel('Y (mm)');
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;

%%
figure(3)
```

```
hold on;
box on
grid minor
% Plotting the quad mesh using patch
hPatch_sec_1_sigY= patch('Faces', sec_1_elements, 'Vertices', mesh_nodes, ...
                         'FaceVertexCData', sec_sigY(1), ...
                         'FaceColor', 'flat', 'EdgeColor', 'black', ...
                         'FaceLighting', 'none');

hPatch_sec_2_sigY= patch('Faces', sec_2_elements, 'Vertices', mesh_nodes, ...
                         'FaceVertexCData', sec_sigY(2), ...
                         'FaceColor', 'flat', 'EdgeColor', 'black', ...
                         'FaceLighting', 'none');

hPatch_sec_3_sigY= patch('Faces', sec_3_elements, 'Vertices', mesh_nodes, ...
                         'FaceVertexCData', sec_sigY(3), ...
                         'FaceColor', 'flat', 'EdgeColor', 'black', ...
                         'FaceLighting', 'none');

hPatch_sec_4_sigY= patch('Faces', sec_4_elements, 'Vertices', mesh_nodes, ...
                         'FaceVertexCData', sec_sigY(4), ...
                         'FaceColor', 'flat', 'EdgeColor', 'black', ...
                         'FaceLighting', 'none');

hPatch_sec_5_sigY= patch('Faces', sec_5_elements, 'Vertices', mesh_nodes, ...
                         'FaceVertexCData', sec_sigY(5), ...
                         'FaceColor', 'flat', 'EdgeColor', 'black', ...
                         'FaceLighting', 'none');

hPatch_sec_6_sigY= patch('Faces', sec_6_elements, 'Vertices', mesh_nodes, ...
                         'FaceVertexCData', sec_sigY(6), ...
                         'FaceColor', 'flat', 'EdgeColor', 'black', ...
                         'FaceLighting', 'none');
```

```
hPatch_sec_7_sigY= patch('Faces', sec_7_elements, 'Vertices', mesh_nodes, ...
                         'FaceVertexCData', sec_sigY(7), ...
                         'FaceColor', 'flat', 'EdgeColor', 'black', ...
                         'FaceLighting', 'none');



% Set axis properties
axis equal;
xlabel('X (mm)');
ylabel('Y (mm)');
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;

% Adding color bar with a label
colorbar;                                % Add color bar
colormap('jet');                         % Optional: Set the color map to 'jet
    ' (can use other colormaps)
c = colorbar;
c.Label.String = 'Yield Strength [MPa]';
c.Label.FontWeight = 'bold';
c.Label.FontSize = 14;

mtrl_use = section_area_quad_mesh(hPatch_sec_1)*infill_rate_sec(1)/100 ...
          +section_area_quad_mesh(hPatch_sec_2)*infill_rate_sec(2)/100 ...
          +section_area_quad_mesh(hPatch_sec_3)*infill_rate_sec(3)/100 ...
          +section_area_quad_mesh(hPatch_sec_4)*infill_rate_sec(4)/100 ...
          +section_area_quad_mesh(hPatch_sec_5)*infill_rate_sec(5)/100 ...
          +section_area_quad_mesh(hPatch_sec_6)*infill_rate_sec(6)/100 ...
          +section_area_quad_mesh(hPatch_sec_7)*infill_rate_sec(7)/100;

```

```
%% Command Window Output:
fprintf('Based on the PLA material properties gathered, we decide that: \n');
fprintf('The best PLA color for this design is %s;\n',PLA_color);
fprintf('Section 1 has the infill rate of %3.0f%%;\n',infill_rate_sec(1));
fprintf('Section 2 has the infill rate of %3.0f%%;\n',infill_rate_sec(2));
fprintf('Section 3 has the infill rate of %3.0f%%;\n',infill_rate_sec(3));
fprintf('Section 4 has the infill rate of %3.0f%%;\n',infill_rate_sec(4));
fprintf('Section 5 has the infill rate of %3.0f%%;\n',infill_rate_sec(5));
fprintf('Section 6 has the infill rate of %3.0f%%;\n',infill_rate_sec(6));
fprintf('Section 7 has the infill rate of %3.0f%%;\n',infill_rate_sec(7));
fprintf('Overall Material usage is %8.2f. (The lower the better)\n',mtrl_use);
```